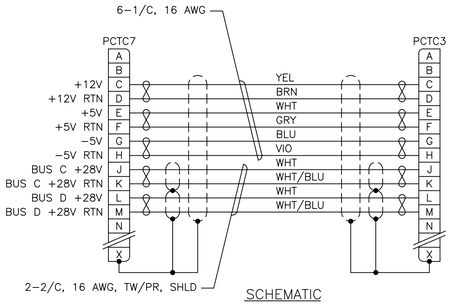
COMMERCIAL PARTS LIST

ITEM	QTY	PART NUMBER	DESCRIPTION	MANUFACTURER
C1	1	D38999/26FH21PB	CONNECTOR, CIRCULAR, PLUG, 21 POS.	QPL
C2	2	380HS115UC2312A3	BACKSHELL, STRAIGHT	GLENAIR
C3	A/R	M39029/58-364	CONTACT, PIN, SIZE 16, 16-20 AWG	QPL
C4	1	D38999/26FH21SB	CONNECTOR, CIRCULAR, PLUG, 21 POS.	QPL
C5	A/R	M39029/56-352	CONTACT, SOCKET, SIZE 16, 16-20 AWG	QPL
C6	A/R	M27500-16RC2S06	2/C, 16 AWG TWISTED, SHIELDED, JACKETED	QPL
C7	A/R	M22759/11-16-4	1/C, 16 AWG, PTFE INSULATED, YELLOW	QPL
C8	A/R	M22759/11-16-1	1/C, 16 AWG, PTFE INSULATED, BROWN	QPL
C9	A/R	M22759/11-16-9	1/C, 16 AWG, PTFE INSULATED, WHITE	QPL
C10	A/R	M22759/11-16-8	1/C, 16 AWG, PTFE INSULATED, GRAY	QPL
C11	A/R	M22759/11-16-6	1/C, 16 AWG, PTFE INSULATED, BLUE	QPL
C12	A/R	M22759/11-16-7	1/C, 16 AWG, PTFE INSULATED, VIOLET	QPL
C13	A/R	2216 B/A GRAY	EPOXY, 2 PART	3M
C14	A/R	THFE4-500015	HEAT SHRINK TUBING, EXPANDED PREHEAT ID 0.500"	SAINT-GOBAIN
C15	A/R	2146	BRAID, TINNED COPPER, 1/2"	ALPHA
C16	A/R	5011001332S	EXPANDABLE SLEEVING, 1/2", BLACK	BENTLEY HARRIS
C17	A/R	TZE-211	LABEL, P-TOUCH, 1/4", BLACK ON WHITE	BROTHER

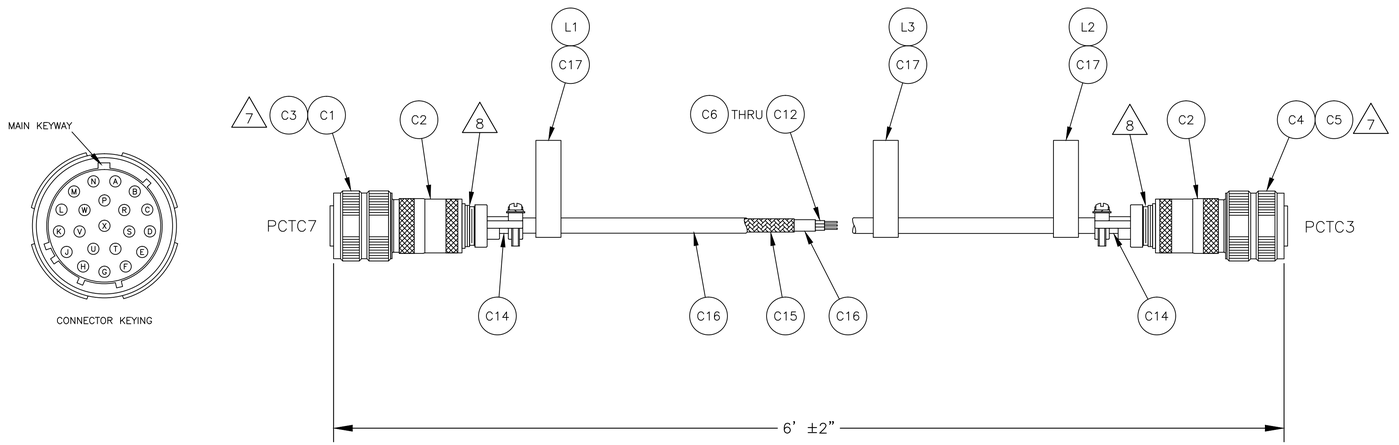
LABEL LIST	
ITEM	INSCRIPTION
L1	PCTC7
L2	PCTC3
L3	S1101EGA4581 SN xxxx



SCHEMATIC

NOTES:

1. MATERIAL CERTIFICATION REQUIRED PER QUALITY LEVEL SPECIFIED IN WORK ORDER/SOW.
2. ESD HANDLING AND CONTROL PER ESTABLISHED PROCEDURES.
3. HANDLE ALL PARTS AND ASSEMBLIES WITH CLEAN, LINT-FREE, ANTI-STATIC GLOVES OR FINGER COTS.
4. HARNESS ASSEMBLY AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.4 AND THE FOLLOWING NOTES:
   a. ALL WIRES 16 AWG UNLESS SPECIFIED.
   b. TO PREVENT POTENTIAL DAMAGE (COLD FLOW) OF THE UNDERLYING WIRE INSULATION, SEPARATOR SHALL BE APPLIED OVER THE WIRE BUNDLE TO GIVE THE WIRE CONTINUOUS PROTECTION.
   c. WIRE TWIST SHALL BE BETWEEN 8 AND 16 TIMES THE OUTER DIAMETER OF THE HARNESS UNLESS OTHERWISE NOTED. (FROM NASA-STD-8739.4 PARA 7.3.20)
5. FOR TENSILE PULLS SEE TENSILE PULL STRENGTH LIST. TENSILE PULL STRENGTHS PER NASA-STD-8739.4 AND/OR THE MANUFACTURER'S SPECIFICATIONS.
6. INSTALL CONTACTS IN ALL UNUSED POSITIONS OF CIRCULAR CONNECTORS.
7. PERFORM CONTACT RETENTION TEST PER NASA-STD-8739.4 AND/OR THE MANUFACTURER'S SPECIFICATIONS. CONTACTS REQUIRING A PULL TEST FOR RETENTION SHALL BE PERFORMED AS FOLLOWS: A GENTLE PULL ON THE WIRE SHALL BE USED TO DETERMINE IF THE CONTACT IS SEATED PROPERLY.
8. TORQUE BACKSHELLS (ITEM C2) TO 80 ±5 IN-LBS. STAKE BACKSHELLS USING EPOXY (ITEM C13).
9. TEST PER NASA-STD-8739.4 PARA 18.2.6 USING THE FOLLOWING:
   CONTINUITY: < 5 OHMS
   DWV: VOLTAGE: 1500 ±75VDC
   CURRENT: < 1 MILLIAMP
   DURATION: < 1 MINUTE
   IR: VOLTAGE: 500 ±75VDC
   RESISTANCE: < 100 MEG OHMS
   DURATION: < 1 MINUTE
10. PLACE COMPLETED ASSEMBLY IN CLEAN, STATIC-SHIELDING PACKAGING. IDENTIFY PART BY TAGGING WITH PART NUMBER AND SERIAL/LOT NUMBER AS SPECIFIED IN WORK ORDER/STATEMENT OF WORK PER SAE AS478, METHOD 35D. STORE IN A PROJECT APPROVED STORAGE AREA.



CRIMP TOOL LIST				
CONTACT P/N	CRIMP TOOL	POSITIONER	I/E TOOL	REF. DES.
M39029/58-364	M22520/1-01	M22520/1-04	M81969/14-03	PCTC7
M39029/56-352	M22520/1-01	M22520/1-04	M81969/14-03	PCTC3

TENSILE PULL STRENGTH LIST			
MANUFACTURER	CONTACT P/N	WIRE AWG	LBS
QPL	M39029/58-364	16	41.0
QPL	M39029/56-352	16	41.0

BACKSHELL FASTENER TORQUE TABLE		
SCREW SIZE	TORQUE RANGE	UNITS
2-56	2 ±0.5	IN-LB
4-40	4 ±0.5	IN-LB
6-32	6 ±1.0	IN-LB
8-32	8 ±1.0	IN-LB
10-32	10 ±1.0	IN-LB
.250-20	12 ±1.0	IN-LB

NEXT ASSEMBLY

UNLESS OTHERWISE SPECIFIED:
1. ALL DIMENSIONS ARE IN INCHES
2. BREAK EDGES .010-.020
3. INSIDE CORNERS R.02 MAX
4. LINEAR TOLERANCES:
   .X ± 0.03
   .XX ± 0.01
   .XXX ± 0.005
5. ANGULAR TOLERANCE ± 0.5°

THIRD ANGLE PROJECTION

THIS DRAWING IS UNDER CONFIGURATION CONTROL: DO NOT REVISE WITHOUT AN APPROVED ECO.

APPROVED: E. McCLOSKEY 6/2/15
PRIOR TO USE, CHECK WITH CONFIGURATION MANAGEMENT FOR ANY OUTSTANDING ECOS.

DESIGNER	M. HOMKO 5/28/15	QUALITY	D. SEELEY 6/1/15
REVIEWER	J. BRUEWER 6/2/15	MATERIALS	M. WILKINSON 6/1/15
MECHANICAL	A. DRAKE 5/28/15	PROJ LEAD	B. BORowski 6/2/15
ELECTRICAL	T. GOBELI 6/1/15		
STRUCTURAL	C. WERNER 6/1/15		
SAFETY	W. BORRELLI 5/29/15		

DIMENSIONS AND TOLERANCING PER ASME Y14.5M-1994

ZIN TECHNOLOGIES
6745 ENGLE ROAD
AIRPORT EXECUTIVE PARK
CLEVELAND, OHIO 44130
(440) 625-2200
www.zin-tech.com

CONTRACT NUMBER: NNC14CA02C
PROJECT CODE: S2-BT3

REV	DESCRIPTION	APPROVED

REVISION HISTORY

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
JOHN H. GLENN RESEARCH CENTER
LEWIS FIELD
CLEVELAND, OHIO

TITLE
PCTC7 EXTENSION CABLE
(VAC OUTSIDE CHAMBER SIDE)

SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101EGA4581	-

SCALE	INITIAL REL DATE	SHEET
NONE	06/02/15	1 OF 1

RELEASE LEVEL: RELEASED
MODEL NAME: S1101EGA4581_REV_-.DWG

AutoCAD 2012

DWG NO S1101EGA4581 SH 1 REV -